Crossdem: Comparing Politicians' Rhetoric With Democracy Indicators

MARCO BELLÒ, University of Padua, Italy

Comparison between politicians' speeches and V-Dem democracy indices. After building a corpus of public speeches from Italian Prime Minister, U.S. Presidents and French Prime Ministers and Presidents, NLP analyses will be performed, from text complexity to hate speech recognition. The ensuing results will be compared with V-Dem democracy indices to shed light on possible correlations. Lastly, a transformer model will be fine-tuned to interpret political language and classify hate speech.

CCS Concepts: • **Computing methodologies → Natural language processing**; • **Social and professional topics → Hate speech**; **Political speech**.

Additional Key Words and Phrases: AI, NLP, Politics, Speeches

ACM Reference Format:

Marco Bellò. 2026. Crossdem: Comparing Politicians' Rhetoric With Democracy Indicators. 1, 1 (May 2026), 14 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Politicians are some of the most influential people in human society: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page (1984) [31] showed how nearly all 20^{th} century U.S. Presidents have tried to lead public opinion, with varying degrees of success. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, speeches play a vital role in the functioning of society [3]: leaders depend on the verbal power to persuade people [9].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et al. [11] evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicating that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office.

Having established the importance of speeches, we can differentiate between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. This last category is especially interesting since it affords politicians a direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57, 000 times between May 2009 and January 2021 alone [27], while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social network interactions in 2025 [38].

Author's address: Marco Bellò, marco.bello.vi@gmail.com, University of Padua, Padua, Italy.

Manuscript submitted to ACM

On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents. Moreover, until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated them into a "language of the parliament" [20, 29].

2 SOCIAL NETWORKS

Broadly speaking, the goal of this project is to gather a corpus of politicians' speeches and perform linguistic analyses on it. The idea is to start from standard NLP tasks such as word frequency and text complexity, to then move to classification tasks such as hate speech recognition, and finally to fine-tuning an LLM.

This objective is why I ultimately decided to gather a corpus of *external speeches* rather than internal ones. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung et al. [10] did a controlled study that scaled the vocabulary of the language model from 24K to 196K while holding data, computation, and optimization unchanged. They discovered that models are disproportionately optimized on the part of language that appears most often, meaning they are way better at understanding common words and patterns. This translates to the fact that internal speeches are less understandable for an AI agent.

As previously stated, external speeches include social network communications. While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [16, 28]. There is evidence of an increase in political participation due to social media usage, as well as risks for the functioning of democracy [2, 26], and there have been attempts to predict election results with social media data analyses. Rita et al. measure sentiment polarity on Twitter, and conclude that tweets' sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions [35].

Opposite results can be seen in Belcastro et al.'s [5] 2022 study: a real-time analysis was carried out during the 2020 US presidential election campaign, correctly identifying the leading candidate before Election Day in 10 out of 11 swing states.

Silva et al. [37] managed to match tweets against parliamentary speeches to measure politicians' sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena.

Lastly, social networks open the possibility of semi-automatized corpus building.

3 PROJECT GOAL

Let's now define the project's goal.

I want to compare politicians' speeches among three countries, namely Italy, France and the United States, between 1945 and 2025, and then cross-compare the resulting analyses with democracy level indicators taken from the V-Dem dataset[1] and fine-tune an LLM to better interpret political speech and classify populism and hate speech.

[1]The V-Dem Dataset is a multidimensional and disaggregated dataset that reflects the complexity of the concept of democracy as a system of rule that goes beyond the simple presence of elections: https://www.v-dem.net/data/the-v-dem-dataset/.

3.1 Comparing Italy, France and the U.S.

Italy was a natural choice being the country I was born and raised in, therefore the one I am most knowledgeable about, but more than that, it has always been a fascinating research case study, being often referred to as an "anomaly" or, by authors such as Marc Lazar, as a kind of *laboratory* of a general change of European democracies. Since the 1990s, the state of Italian democracy has raised numerous questions, stimulating the curiosity of the international media, causing heated debates, providing material for seminars and conferences, and engendering academic and journalistic publications by Italians and foreigners alike. Lazar's opinion is that Italy is "at the same time an indicator of a more general democratic malaise and of the multiple attempts of invention of a new democratic order that are emerging all over the European Union". Similar phenomena affect, in their own ways, other European democracies, making it a meaningful case study [24].

Italian, French, and American histories intersected often during the past century: being at times allies, at times enemies during the Second World War, in the ensuing Cold War the United States encountered unexpected challenges from the Italian and French Communist Parties, the two strongest and most anti-American in Western Europe. Brogi [6] argues that the resistance to Americanization was necessary to legitimize the French and Italian Communist Parties' own existence.

Moreover, each country illustrates a different approach to democratic governance. They use three different executive models: the U.S. opted for a full presidential, where the President holds all power, France uses a semi-presidential system where power is split between the President and the Prime Minister, and Italy a Parliamentary one where the Prime Minister holds the power but depends on the Parliament. The party system is also different, since France and Italy have multi-party systems while the U.S. is strongly bipartisan, being divided between Democrats and Republicans. Lastly, the state system employed is also quite different: France's state is unitary and vertical, while Italy's regions hold strong power and independence. The U.S., on the other hand, is a Federation.

3.2 Roadmap

The first step is building a corpus of politicians' speeches, one corpus for Italian speeches, one for French speeches and one for American speeches. To limit the scope, I decided to take only speeches from Italian Prime Ministers, U.S. Presidents, and either French Presidents when the President and the Prime Minister come from the same party, or French Prime Ministers when the President and Prime Minister are in opposition to each other. This is justified by the fact that when they are politically aligned, the President dominates. During cohabitation (opposing political camps), the Prime Minister gains real policy autonomy [30].

Speeches can be taken from different sources, such as official speeches, interviews, or social network posts, and can be in different media formats, such as textual, audio or video. For the latter two, a translating tool will be needed (such as a speech-to-text library).

The second step is using the corpus to perform linguistic analyses. The corpus can be subjected to data augmentation[2], since we can expect the density of speeches to decrease as we go back in time. Then, the corpus will be pre-processed, with procedures such as tokenization, stemming and lemmatization. We can then compute word frequency and text complexity measures, as well as hate speech and populism identification.

[2]In its most general sense, data augmentation denotes methods for supplementing so-called incomplete datasets by providing missing data points in order to increase the dataset's analyzability [22].

The third step is cross-referencing the linguistic metrics with democratic indicators such as freedom of expression, freedom of association, transparency and fairness of law and other such metrics. All democracy metrics can be found in the V-Dem dataset.

The fourth step is to fine-tune a pre-trained transformer model, for example Mistral 7B[3], to make it better understand political speeches in three different languages, and to train it in a classification task for hate speech and populism recognition.

4 BUILDING THE CORPUS

The natural first step was to look at other corpora that aggregate similar data.

The **Europarl** dataset [23] aggregates proceedings of the European Parliament in twenty-one different European languages[4] with the goal of generating sentence-aligned text for statistical machine learning translation systems. The overall structure is akin to a collection of plaintext with minimal tagging.

ItaParlCorpus [13], on the other hand, is a comprehensive, annotated, and machine-readable database of Italy's parliamentary speeches spanning from 1948 to 2022. Each speech is identified by a precise date, the role of the speaker, their party name and family, and the legislature number. It is not, however, very rich in terms of linguistic processing tagging.

IMPAQTS [12] is a multimodal corpus of around 2.65 million tokens, including 1,500 speeches uttered by 150 prominent politicians, spanning from 1946 to 2023. For each speaker, the corpus contains 4 parliamentary speeches, 2 rallies, 1 party assembly, and 3 statements (in person or broadcasted). The goal is to annotate political rhetoric that hides a deeper meaning, for example by employing sarcasm. There are four main types of annotation: implication (conventional, generalized, etc.), presupposition (e.g., pragmatic), vagueness (syntactic, semantic, metaphoric), and topicalization (either syntactic or prosodic).

Parola di Leader, and specifically the corpus LP4 [17], is a corpus of more than five million occurrences taken from the stenographic archives of the Chamber of Deputies of Italy[5], between 1948 (first legislature of the Italian Republic) and 2011 (excluding the Monti Government, the 61^{st} legislature).

They selected a list of thirty "highly influential politicians"[6] and then for each of them they selected some speeches. Overall, they collected 1877 distinct speeches.

The corpus could be considered a collection of thirty distinct corpora, one for each politician. Each of these has, among others, the following features: name of the politician, role of the politician (for example, Prime Minister), whether the politician is in the majority party or in the opposition, legislature number and sitting number of the day, date of the sitting, topic of the speech, name of the current Government (e.g., *Berlusconi I*), and lastly a reference to the file that holds the actual speech.

[3]Mistral 7B the most powerful language model for its size to date: https://mistral.ai/news/announcing-mistral-7b.
[4]All 21 Europarl languages: Romanic (French, Italian, Spanish, Portuguese, Romanian), Germanic (English, Dutch, German, Danish, Swedish), Slavic (Bulgarian, Czech, Polish, Slovak, Slovene), Finno-Ugric (Finnish, Hungarian, Estonian), Baltic (Latvian, Lithuanian), and Greek.
[5]Chamber of Deputies official online public archive: https://legislatureprecedenti.camera.it/.
[6]The full list of politicians follows: Giorgio Almirante, Giuliano Amato, Giulio Andreotti, Enrico Berlinguer, Silvio Berlusconi, Pier Luigi Bersani, Fausto Bertinotti, Rosy Bindi, Emma Bonino, Umberto Bossi, Pier Ferdinando Casini, Francesco Cossiga, Bettino Craxi, Massimo D'Alema, Alcide De Gasperi, Ciriaco De Mita, Antonio Di Pietro, Amintore Fanfani, Gianfranco Fini, Ugo La Malfa, Aldo Moro, Pietro Nenni, Achille Occhetto, Marco Pannella, Romano Prodi, Giuseppe Saragat, Giovanni Spadolini, Palmiro Togliatti, Valter Veltroni, Nichi Vendola.

There is also a corpus for the vocabulary that holds, for each word, the total number of occurrences, a part-of-speech[7] tag, its lemma, a count for the number of occurrences for each politician, majority party and opposition, and lastly the imprinting tag (e.g., masculine singular).

Lastly, **ParlaMint** [14, 15] is a compiled collection of comparable parliamentary corpora for a number of countries and languages. ParlaMint corpora are interoperable, i.e. encoded to a very constrained common ParlaMint schema, a specialization of the Parla-CLARIN[8] recommendations, which are a customization of the TEI Guidelines[9].

It is a collection of parliamentary sittings split into utterances. An example of two utterances could be *Politician: "I ask permission to speak.", President of the Senate: "Permission granted."*. For each utterance there are, among others, the following metadata: an ID for both the whole sitting and the utterance itself, a title of the sitting, a date of the sitting, the State Body where it took place (e.g., the Senate of the Republic), the name of the speaker and their affiliation, as well as the name of their coalition.

Speeches are collected before being split into their own datasets, and for each speech we have, among others, a unique ID and a date, the type of speech (e.g., *Interview* or *Rally*), the venue channel (e.g., *BBC News*), the unique ID of the speaker and their role (e.g., *host* or *main speaker*), the text of the speech itself and its type (e.g., *Question*, *Answer*, *Interruption*) and whether there have been any *Incidents* and their types.

Incidents are breaks in the flow of the conversation, and they can be of different types[10], for example *"leaving"* or *"break"*. If an incident is of the kinesic type, it can be further subdivided into different kinesic types, such as *"applause"* or *"laughter"*[11]. If the incident is of the vocal type, it can be further subdivided into different vocal types, such as *"question"* or *"shouting"*[12].

Speakers and Parties also have their own datasets, with unique IDs, names, affiliations, roles and the like.

Overall, the taxonomy of ParlaMint is strongly codified, which makes the dataset machine-readable and easier to interpret. All data is also saved in XML format.

4.1 My Corpus

Ultimately, I did not manage to find any corpus that I could directly integrate or use for my purposes, but I took inspiration from ParlaMint for the overall structure. At the time of writing, the corpus is not yet compiled, so the structure may change. The structure for the Italian corpus is the only one present at the moment. The structure is explained as tables and columns (or *features*) of said tables.

Italian Government Table:

- Column *government*:
  - e.g., `degasperi2`: "De Gasperi II";
  - e.g., `spadolini1`: "Spadolini I";
  - e.g., `forlani`: "Forlani";
- Column *primeMinister*:
  - e.g., `degasperi`: `prime_ministers.id==degasperi`;
- Column *dateStart*: ISO 8601 (YYYY-MM-DD);

[7] In grammar, a part of speech or part-of-speech (abbreviated as POS or PoS) is a category of words (or, more generally, of lexical items) that have similar grammatical properties [32].
[8] Parla-CLARIN, a TEI schema for corpora of parliamentary proceedings: https://clarin-eric.github.io/parla-clarin/.
[9] P5 TEI Guidelines: https://tei-c.org/guidelines/p5/.
[10] Full list of incident types: "action", "incident", "leaving", "entering", "break", "pause", "sound", "editorial".
[11] Full list of kinesic types: "kinesic", "applause", "ringing", "signal", "playback", "gesture", "smiling", "laughter", "snapping", "noise".
[12] Full list of vocal types: "greeting", "question", "clarification", "speaking", "interruption", "exclamat", "laughter", "shouting", "murmuring", "noise", "signal".

- Column *dateEnd*: ISO 8601 (YYYY-MM-DD);
- Column *legislature*:
  - e.g., 1: "I Legislatura";
  - e.g., 15: "XV Legislatura";
- Column *pm_party*: The political party of the Prime Minister;
- Column *gov_leaning*: The political leaning of the specific government.

Prime Ministers Table:

- Column *pm_ID*:
  - e.g., `degasperi`;
  - e.g., `draghi`;
  - e.g., `meloni`;
- Column *pm_name*:
  - e.g., `Alcide`;
  - e.g., `Mario`;
  - e.g., `Giorgia`;
- Column *pm_familyName*:
  - e.g., `De Gasperi`;
  - e.g., `Draghi`;
  - e.g., `Meloni`;
- Column *pm_birthDate*: The birth date of the Prime Minister;
- Column *pm_deathDate*: The death date of the Prime Minister.

The dataset will have a structured taxonomy; for example, the political leaning will be codified as follows:

- `mixed`: e.g., CLN[13], Pentapartito[14];
- `RR`: Far-Right;
- `LL`: Far-Left;
- `right`: Right;
- `left`: Left;
- `center`: Center;
- `centerRight`: Center-Right;
- `centerLeft`: Center-Left.

4.2 Data Augmentation and Synthetic Data

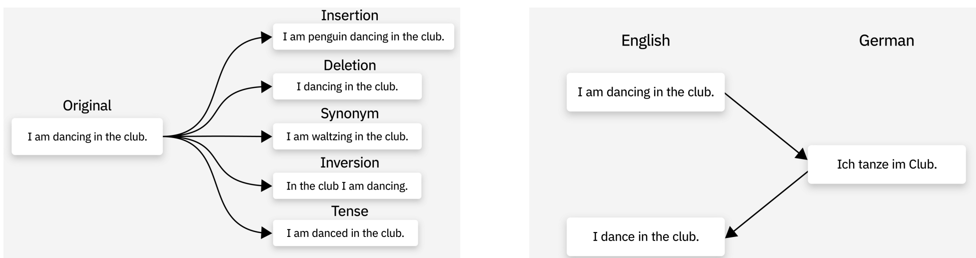
It is likely that for older politicians there are fewer speeches available, i.e., the dataset becomes more sparse as we go back in time. Moreover, speeches will be in three languages, namely Italian, French and English. For this reason, data augmentation techniques may be useful (figure 1).

[13]The National Liberation Committee (Italian: Comitato di Liberazione Nazionale, CLN) was a political umbrella organization and the main representative of the Italian resistance movement fighting against the occupying forces of Nazi Germany and the fascist collaborationist forces of the Italian Social Republic. It was composed of Christian Democrats, Socialists, Communists, and Liberals.

[14]The Pentapartito was the coalition government of five Italian political parties that formed between June 1981 and April 1991. It was composed of Christian Democrats, Social Democrats, Liberal Democrats, and Conservative Liberals.

Data augmentation uses pre-existing data to create new data samples that supplement so-called incomplete datasets by providing missing data points, improving machine learning model optimization and generalization. In other words, data augmentation can reduce overfitting and improve model robustness. This is different from generating synthetic data, which is entirely artificial: data augmentation copies existing data and transforms those copies to increase the diversity and amount of data in a given set.

Rule-based approaches include relatively simple find-and-replace techniques, such as random deletion or insertion. In this strategy, one or more words in a string are replaced with their respective synonyms as recorded in a predefined thesaurus, such as WordNet or the Paraphrase Database. Back-translation (figure 1b) is a technique where the corpus is translated into another language before being translated back to the original language [22].



(a) Increasing data density.

(b) Back-translation.

Fig. 1. Data augmentation techniques, such as synonym replacement and back-translation. Source: Kavlakoglu, IBM [22].

Synthetic data is inherently privatized, since it cannot be ascribed to any real person. This makes it free from any GDPR control system and privacy concerns. It can be of two types: fully synthetic, which does not include any real-world information, or partially synthetic, where portions of the original dataset are replaced with artificial values. Usually, it is generated with either GANs or transformer models and comes with two main problems: bias and model collapse. Synthetic data can exhibit biases present in the real-world data that the model is based on, and when an AI model is repeatedly trained on AI-generated data, its performance declines quickly [7, 21, 34].

We probably will not need the enhanced privacy granted by synthetic data since the dataset will be compiled using public speeches from public figures.

5 LINGUISTIC ANALYSIS

Sawicki et al. 2023's survey [36] tells us that English is by far the most represented language in the field of natural language processing. Fortunately, French and Italian are still in the top eight (figure 2), so finding libraries already capable of handling them should be easy enough. It is worth noting that results could be biased since the authors looked for papers written in English.

The pipeline for pre-processing is quite standard and involves, among other steps, removing stop words, part-of-speech tagging, tokenization and lemmatization, as well as building text embeddings.

FastText[15] is an open-source, free, lightweight library that allows users to learn text representations and text classifiers; it gained significant popularity due to its performance in computing text embeddings.

[15]FastText is a lightweight library for efficient text classification and representation learning: https://fasttext.cc/.

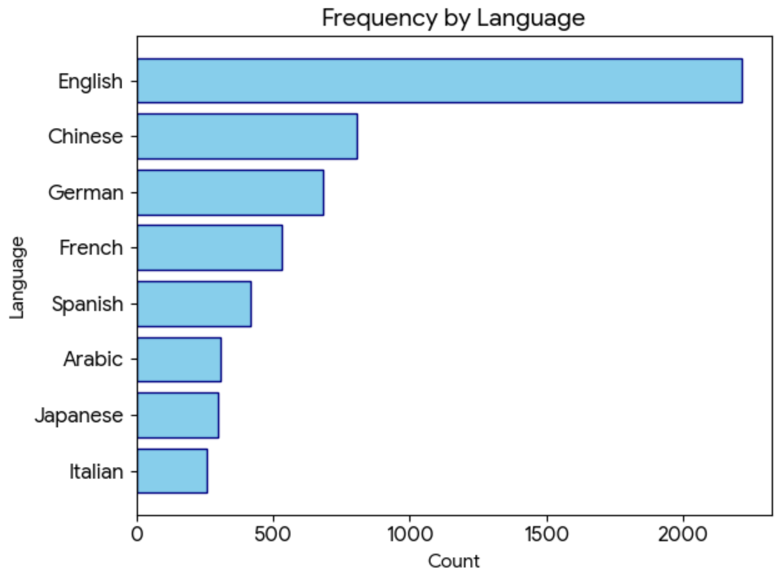


Fig. 2. Most popular languages in the field of Natural Language Processing [36]

LLM2Vec [4] is a recipe to convert decoder-only LLMs into text encoders so that they can be used to generate state-of-the-art text embeddings.

Stanza[16] is a collection of accurate and efficient tools for the linguistic analysis of many human languages. Starting from raw text, Stanza divides it into sentences and words, and can then recognize parts of speech and entities, perform syntactic analysis, and more. Stanza brings state-of-the-art NLP models to the languages of your choosing.

spaCy[17] is another well-known and powerful NLP library for Python that supports multiple languages. It also supports tokenization, part-of-speech tagging, dependency parsing, sentence segmentation, text classification, and lemmatization. It supports multi-task learning with pretrained transformers, includes pretrained word vectors, supports custom models in PyTorch and TensorFlow, and has built-in syntax visualizers.

simplemma[18] is also worth mentioning as a lightweight Python lemmatizer.

textacy[19] is a Python library for performing a variety of natural language processing (NLP) tasks. It is built on spaCy and intended to be used both before (pre-processing step) and after (NLP step) it. It assesses text complexity with the Flesch-Kincaid readability test and text diversity with the Type-Token Ratio.

Text complexity can also be measured with **TRUNAJOD**[20], another spaCy-based Python library that implements synonym overlap, coherence between sentences, and other metrics.

[16] Stanza – A Python NLP Package for Many Human Languages: https://stanfordnlp.github.io/stanza/index.html.
[17] spaCy, Industrial-Strength Natural Language Processing: https://spacy.io/.
[18] simplemma is a lightweight toolkit for multilingual lemmatization and language detection: https://pypi.org/project/simplemma/.
[19] textacy: NLP, before and after spaCy. https://pypi.org/project/textacy/.
[20] TRUNAJOD: A text complexity library for text analysis built on spaCy. https://trunajod20.readthedocs.io/en/latest/.

Regarding populism and hate speech classification, there are many transformer-based models available from **Hugging Face**[21]. Some examples follow:

- English: https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-english;
- French: https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-french;
- Italian: https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-italian.

There are transformer models based on RoBERTa for both hate speech and populism classification. For example, the following is a 0.4 billion parameter model based on Trump speeches: https://huggingface.co/coastalcph/roberta-large-ft-trump-populism.

Hate speech can also be detected with the Python library **HateSonar**[22], which uses a BERT-based model.

5.1 Speech To Text

Since there are plenty of video and audio speeches publicly available, they could be integrated into the corpus after transcribing them into text. This can be achieved with libraries such as **SpeechRecognition**[23], which supports several different engines and APIs both online and offline, or OpenAI's **Whisper**[24] library. Whisper is a Transformer sequence-to-sequence model trained on various speech processing tasks, including multilingual speech recognition, speech translation, spoken language identification, and voice activity detection. It comes in six different model sizes; the "small" model requires approximately 2GB of VRAM, making it easy to run locally.

Whisper's performance varies depending on the language. Italian is the fourth best-performing language with a WER[25] of 5.5%. English follows at 9.3% and French at 10.8% [33]. The worst-performing language is Albanian with a WER of 55.7%.

Whisper can be used both in Python scripts and via a command-line interface.

6 DEMOCRACY INDEX

There are plenty of sources for measuring democracy levels. Our World in Data mainly uses six: Varieties of Democracy (V-Dem), the Lexical Index of Electoral Democracy (LIED) by Skaaning et al. (2015), Freedom House's (FH) Freedom in the World index, the Bertelsmann Transformation Index (BTI) by the Bertelsmann Foundation, the Economist Intelligence Unit's (EIU) Democracy Index, and Polity by the Center for Systemic Peace [18].

Out of the six mentioned datasets, only V-Dem and LIED have data spanning between 1946 and 2025. However, V-Dem offers more granularity and is generally considered a more complete and robust dataset. It is the best choice if we are interested in both large and small differences in varieties of democracy far into the past, or if we want to use country experts to measure characteristics of political systems that are difficult to observe. These experts are anonymous and are primarily academics or members of the media and civil society. They are also often nationals or residents of the country they assess; therefore, they know its political system well and can evaluate aspects that are difficult to observe. V-Dem's own team of researchers supplements these expert evaluations [19].

---

[21] Hugging Face, Inc., is an American company based in New York City that develops computational tools for building applications using machine learning. Its transformers library, built for natural language processing applications, and its platform allow users to share machine learning models and datasets and showcase their work. https://huggingface.co/

[22] HateSonar: Hate Speech Detection. https://pypi.org/project/hatesonar/.

[23] SpeechRecognition, a Python library for performing speech recognition with support for several engines and APIs, online and offline. https://pypi.org/project/SpeechRecognition/.

[24] Whisper is a general-purpose speech recognition model. https://github.com/openai/whisper.

[25] Word Error Rate (WER) is a common metric for the performance of speech recognition or machine translation systems. It is measured as an error rate percentage for compared sentences.

6.1 Varieties of Democracy Dataset

All data is available for download at https://v-dem.net/data/the-v-dem-dataset/ or can be accessed directly as an R package.

V-Dem distinguishes between *indicators*—lower-level data that is often not normalized—and *indices*, which aggregate indicators and are normalized on a scale between zero and one.

There are four datasets: *Coder-Level*, with raw data and 273 indicators intended for the experts; *CD Country-Date*, with day-to-day granularity, 531 indicators, and 95 aggregated indices; *CY Country-Year*, with year-level granularity, 5 indices, 93 sub-indices, and 179 indicators; and lastly, *CY Full + Others*, with year-level granularity, 531 indicators, 251 indices, and 62 external indicators.

I must use the *CD Country-Date* dataset (version 16, released in March 2026) since, to keep track of Italian governments, year-level granularity is not enough. This is evident from Figure 3: half of Italy's governments lasted less than a year.

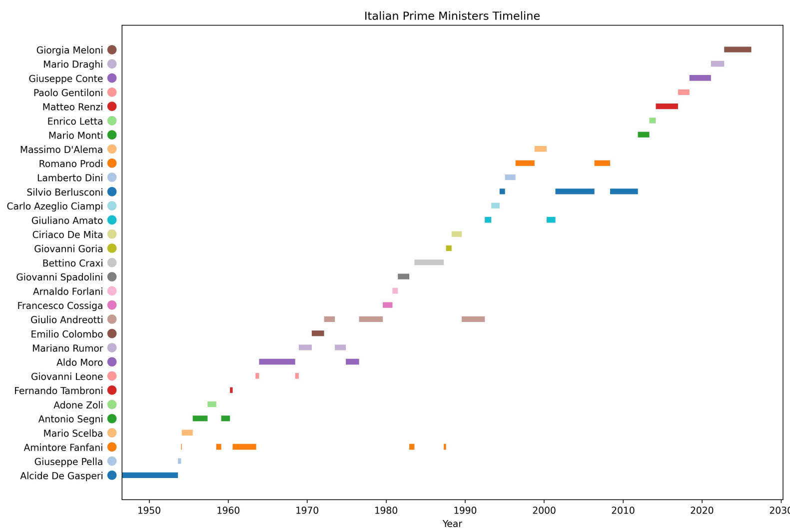


Fig. 3. Italian Prime Ministers for each Italian Republic, from the first (1946) to the latest (2025).

The dataset covers Italy (ID 82) between 1861 and 2025, France (ID 76) between 1789 and 2025, and the U.S. (ID 20) between 1789 and 2025.

Useful identifiers include: `country_id`, unique for each country; `country_text_id`, ISO 3166-1 alpha-2 country codes (e.g., *IT*, *FR*); `historical_date`, in *YYYY-MM-DD* format, used for election-date-specific variables (all other variables use January 1st); and `gapstart` and `gapend`, representing time periods where no data is available.

Democracy Indices are the highest-level indices, each normalized to a scale between zero and one unless otherwise specified: `v2x_polyarchy`, the *electoral* democracy index (aggregates `v2x_elecoff`, `v2xel_frefair`, `v2x_frassoc_thick`, `v2x_suffr`, `v2x_freexp_altinf`); `v2x_libdem`, the *liberal* democracy index (aggregates `v2x_polyarchy`, `v2x_liberal`); `v2x_partipdem`, the *participatory* democracy index (aggregates `v2x_polyarchy`, `v2x_partip`); `v2x_delibdem`, the *deliberative* democracy index focusing on process, respectful dialogue, and the common good (aggregates `v2x_polyarchy`, `v2xdl_delib`); and `v2x_egaldem`, the *egalitarian* democracy index considering social status, inequalities, and resource distribution (aggregates `v2x_polyarchy`, `v2x_egal`).

Components of Democracy Indices consist of second-level indices and subcomponents: `v2x_freexp_altinf`, freedom of expression and alternative information (aggregates *v2mecenefm*, *v2meharjrn*, *v2meslfcen*, *v2xcl_disc*, *v2clacfree*, *v2mebias*, *v2mecrit*, *v2merange*); `v2x_frassoc_thick`, freedom of association for parties and civil society organizations (aggregates *v2psparban*, *v2psbars*, *v2psoppaut*, *v2elmulpar*, *v2cseeorgs*, *v2csreprss*, `v2x_elecreg`); `v2x_suffr`, the percentage of adult suffrage (aggregates *v2elsuffrage*); `v2xel_frefair`, the quality of free and fair elections (aggregates *v2elembaut*, *v2elembcap*, *v2elrgstry*, *v2elvotbuy*, *v2elirreg*, *v2elintim*, *v2elpeace*, *v2elfrfair*, `v2x_elecreg`); `v2xcl_rol`, rule of law, transparency, and fair enforcement (aggregates *v2clrspct*, *v2cltrnslw*, *v2cltort*, *v2clkill*, *v2clrelig*, *v2clfmove*, `v2xcl_dmove`, `v2xcl_slave`, `v2xcl_acjst`, `v2xcl_prpty`); `v2x_jucon`, executive law compliance and judiciary independence (aggregates *v2exrescon*, *v2jucomp*, *v2juhccomp*, *v2juhcind*, *v2juncind*); `v2xlg_legcon`, executive external scrutiny (aggregates *v2lgqstexp*, *v2lgotovst*, *v2lginvstp*, *v2lgoppart*); `v2x_partip`, individual participation (aggregates `v2x_cspart`, `v2xdd_dd`, `v2xel_locelec`, `v2xel_regelec`); `v2x_cspart`, participation in associations like labor unions or NGOs (aggregates *v2pscnslnl*, *v2cscnsult*, *v2csprtcpt*, *v2csgender*); `v2xdd_dd`, the direct vote index for referendums and ballots (aggregates *v2ddlexci* through *v2ddthreci*); `v2xel_regelec` and `v2xel_locelec`, regional and local government elections; `v2xeg_eqprotec`, equal rights across social groups; `v2xeg_eqaccess`, equality of accessing power; and `v2xeg_eqdr`, equality of resource distribution (aggregates *v2dlencmps*, *v2dlunivl*, *v2peedueq*, *v2pehealth*).

7 MODEL FINE-TUNING

Classification tasks can be performed, broadly speaking, in two ways: the first leverages common machine learning and statistical tools, such as logistic regression; the second approach uses Transformer models (commonly referred to today as "AI"). A traditional approach might involve using TF-IDF and logistic regression with the Scikit-Learn library (example code at 1).

Listing 1. Scikit-Learn Logistic Regression for classification task

```python
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline

X_train, X_test, y_train, y_test = train_test_split(df['text'], df['label'], test_size=0.2)

text_clf = Pipeline([
    ('tfidf', TfidfVectorizer(stop_words='english', ngram_range=(1, 2))),
    ('clf', LogisticRegression())
])
```

On the other hand, Transformers are far superior in terms of accuracy for textual analysis, although they are much more resource-demanding. An example would be using Hugging Face's `transformers` library to fine-tune a pre-trained model (example code at 2).

Listing 2. Hugging Face Trainer for fine-tuning

```python
from transformers import AutoModelForSequenceClassification, AutoTokenizer, Trainer, TrainingArguments

trainer = Trainer(
    model=AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=2),
    args=training_args,
    train_dataset=tokenized_datasets["train"],
    eval_dataset=tokenized_datasets["test"],
```

```
)

trainer.train()
```

Regarding the literature on the subject, Card et al. [8] fine-tuned a pre-trained RoBERTa model on congressional speeches in a self-supervised fashion and then further fine-tuned it as a classifier using annotated examples. They achieved 65% accuracy in classifying Congressional Record entries as being pro-immigration, anti-immigration, or neutral.

Albladi et al. [1] surveyed the field of NLP sentiment analysis and found that RoBERTa consistently outperforms other methods in multi-classification tasks. Their results revealed a clear correlation between dataset size and improved model performance, with larger datasets significantly increasing both accuracy and F1 scores across all models. For example, GPT-3.5 reached accuracies of 0.9694, 0.9784, 0.9797, and 0.9875 for various dataset sizes (e.g., 500, 1000, and 2500). However, LLaMA-2 13B emerged as the most balanced option for NLP tasks, striking an optimal balance between computational efficiency, performance, and adaptability. This model can be fine-tuned on freely available GPUs, such as the T4 on Google Colab, using techniques like QLoRA, which makes it an attractive choice for researchers and developers.

Li et al. [25] found that LS-LLaMA substantially outperforms LLMs ten times its size in text classification tasks and demonstrates consistent improvements compared to robust baselines like BERT-Large and RoBERTa-Large.

Considering all of this, it would be interesting to fine-tune a Mistral 7B model, which is claimed to perform better than LLaMA 7B and LLaMA 13B across all benchmarks (Figure 4).

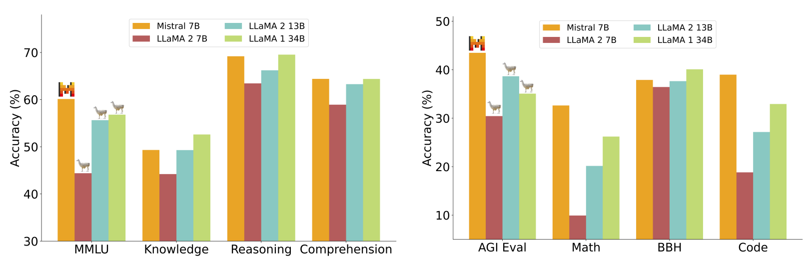


Fig. 4. Mistral 7B consistently outperforms LLaMA on all tasks. Source: Mistral.

8 FEASIBILITY

The project can be completed within the three-year duration of the PhD program. One year is expected to be dedicated to completing the dataset, leaving sufficient time to conduct the ensuing textual analyses.

Furthermore, the majority of the work required for the Italian corpora will be completed in the coming months as part of my Master's thesis project.

This work will tie neatly to my career progression since I changed my specialization from cybersecurity to a more data analysis-oriented path. I also discovered an interest in AI, and specifically natural language processing through transformer models, which this PhD will allow me to explore.

REFERENCES

[1] Aish Albladi, Minarul Islam, and Cheryl Seals. 2025. Sentiment Analysis of Twitter Data Using NLP Models: A Comprehensive Review. *IEEE Access* 13 (2025), 30444–30468. https://doi.org/10.1109/ACCESS.2025.3541494

[2] Eran Amsalem and Alon Zoizner. 2023. Do people learn about politics on social media? A meta-analysis of 76 studies. *Journal of Communication* 73, 1 (Feb. 2023), 3–13. https://doi.org/10.1093/joc/jqac034

[3] Adrian Beard. 2000. *The Language of Politics*. Routledge. https://newuniversityinexileconsortium.org/wp-content/uploads/2022/02/the-language-of-politics.pdf Google-Books-ID: XpqSmh1pjsEC.

[4] Parishad BehnamGhader, Vaibhav Adlakha, Marius Mosbach, Dzmitry Bahdanau, Nicolas Chapados, and Siva Reddy. 2024. LLM2Vec: Large Language Models Are Secretly Powerful Text Encoders. https://doi.org/10.48550/arXiv.2404.05961 arXiv:2404.05961 [cs].

[5] Loris Belcastro, Francesco Branda, Riccardo Cantini, Fabrizio Marozzo, Domenico Talia, and Paolo Trunfio. 2022. Analyzing voter behavior on social media during the 2020 US presidential election campaign. *Social Network Analysis and Mining* 12, 1 (July 2022), 83. https://doi.org/10.1007/s13278-022-00913-9

[6] Alessandro Brogi. 2011. *Confronting America: The Cold War between the United States and the Communists in France and Italy*. The University of North Carolina Press, Chapel Hill. https://muse.jhu.edu/pub/12/monograph/book/21196

[7] Rina Diane Caballar. 2023. What Is Synthetic Data? | IBM. https://www.ibm.com/think/topics/synthetic-data

[8] Dallas Card, Serina Chang, Chris Becker, Julia Mendelsohn, Rob Voigt, Leah Boustan, Ran Abramitzky, and Dan Jurafsky. 2022. Computational analysis of 140 years of US political speeches reveals more positive but increasingly polarized framing of immigration. *Proceedings of the National Academy of Sciences* 119, 31 (Aug. 2022), e2120510119. https://doi.org/10.1073/pnas.2120510119

[9] Jonathan Charteris-Black. 2011. Persuasion, Speech Making and Rhetoric. In *Politicians and Rhetoric: The Persuasive Power of Metaphor*, Jonathan Charteris-Black (Ed.). Palgrave Macmillan UK, London, 1–27. https://doi.org/10.1057/9780230319899_1

[10] Woojin Chung and Jeonghoon Kim. 2025. Exploiting Vocabulary Frequency Imbalance in Language Model Pre-training. https://doi.org/10.48550/arXiv.2508.15390 arXiv:2508.15390 [cs].

[11] Katherine Clayton, Nicholas T. Davis, Brendan Nyhan, Ethan Porter, Timothy J. Ryan, and Thomas J. Wood. 2021. Elite rhetoric can undermine democratic norms. *Proceedings of the National Academy of Sciences* 118, 23 (June 2021), e2024125118. https://doi.org/10.1073/pnas.2024125118

[12] Federica Cominetti, Lorenzo Gregori, Edoardo Lombardi Vallauri, and Alessandro Panunzi. 2024. IMPAQTS: a multimodal corpus of parliamentary and other political speeches in Italy (1946-2023), annotated with implicit strategies. In *Proceedings of the IV Workshop on Creating, Analysing, and Increasing Accessibility of Parliamentary Corpora (ParlaCLARIN) @ LREC-COLING 2024*, Darja Fiser, Maria Eskevich, and David Bordon (Eds.). ELRA and ICCL, Torino, Italia, 101–109. https://aclanthology.org/2024.parlaclarin-1.15/

[13] Joshua Cova. 2025. A new database for Italian parliamentary speeches: introducing the *ItaParlCorpus* dataset. *Italian Political Science Review/Rivista Italiana di Scienza Politica* 55, 1 (March 2025), 77–86. https://doi.org/10.1017/ipo.2025.6

[14] Tomaž Erjavec, Matyáš Kopp, Nikola Ljubešić, Taja Kuzman, Paul Rayson, Petya Osenova, Maciej Ogrodniczuk, Çağrı Çöltekin, Danijel Koržinek, Katja Meden, Jure Skubic, Peter Rupnik, Tommaso Agnoloni, José Aires, Starkaður Barkarson, Roberto Bartolini, Núria Bel, María Calzada Pérez, Roberts Darģis, Sascha Diwersy, Maria Gavriilidou, Ruben van Heusden, Mikel Iruskieta, Neeme Kahusk, Anna Kryvenko, Noémi Ligeti-Nagy, Carmen Magariños, Martin Mölder, Costanza Navarretta, Kiril Simov, Lars Magne Tungland, Jouni Tuominen, John Vidler, Adina Ioana Vladu, Tanja Wissik, Väinö Yrjänäinen, and Darja Fišer. 2025. ParlaMint II: advancing comparable parliamentary corpora across Europe. *Language Resources and Evaluation* 59, 3 (Sept. 2025), 2071–2102. https://doi.org/10.1007/s10579-024-09798-w

[15] Tomaž Erjavec, Maciej Ogrodniczuk, Petya Osenova, Nikola Ljubešić, Kiril Simov, Andrej Pančur, Michał Rudolf, Matyáš Kopp, Starkaður Barkarson, Steinþór Steingrímsson, Çağrı Çöltekin, Jesse de Does, Katrien Depuydt, Tommaso Agnoloni, Giulia Venturi, María Calzada Pérez, Luciana D. de Macedo, Costanza Navarretta, Giancarlo Luxardo, Matthew Coole, Paul Rayson, Vaidas Morkevičius, Tomas Krilavičius, Roberts Darǵis, Orsolya Ring, Ruben van Heusden, Maarten Marx, and Darja Fišer. 2023. The ParlaMint corpora of parliamentary proceedings. *Language Resources and Evaluation* 57, 1 (March 2023), 415–448. https://doi.org/10.1007/s10579-021-09574-0

[16] Eurobarometer. 2023. *Media & News Survey 2023 - November 2023 - - Eurobarometer survey*. Technical Report. European Union. https://doi.org/10.2861/11595

[17] Luca Giuliano. 2019. Il corpus LP. https://www.paroladileader.com/p/blog-page.html

[18] Bastian Herre. 2025. Democracy data: how sources differ and when to use which one. *Our World in Data* (June 2025). https://ourworldindata.org/democracies-measurement

[19] Bastian Herre. 2025. The 'Varieties of Democracy' data: how do researchers measure democracy? *Our World in Data* (April 2025). https://ourworldindata.org/vdem-electoral-democracy-data

[20] Günter Holtus and Michele Cortelazzo. 1985. Dal parlato al (tra)scritto: i resoconti stenografici dei discorsi parlamentari. In *Gesprochenes Italienisch in Geschichte und Gegenwart*. G. Narr, 86–118. https://www.uni-saarland.de/fileadmin/upload/fachrichtung/romanistik/W._Schweickard/Publikationen/Gesprochenes_Italienisch_in_Geschichte_und_Gegenwart.pdf Google-Books-ID: NfwlAAAAMAAJ.

[21] info@gdpr advisor.com. 2025. GDPR and Synthetic Data: Balancing Privacy and AI Model Training. https://www.gdpr-advisor.com/gdpr-and-synthetic-data-balancing-privacy-and-ai-model-training/

[22] Jacob Murel Ph D. Kavlakoglu, Eda. 2024. What is data augmentation? | IBM. https://www.ibm.com/think/topics/data-augmentation

[23] Philipp Koehn. 2005. Europarl: A Parallel Corpus for Statistical Machine Translation. In *Proceedings of Machine Translation Summit X: Papers*. Phuket, Thailand, 79–86. https://aclanthology.org/2005.mtsummit-papers.11/

[24] Marc Lazar. 2013. Testing Italian democracy. *Comparative European Politics* 11, 3 (May 2013), 317–336. https://doi.org/10.1057/cep.2012.40

[25] Zongxi Li, Xianming Li, Yuzhang Liu, Haoran Xie, Jing Li, Fu-lee Wang, Qing Li, and Xiaoqin Zhong. 2023. Label Supervised LLaMA Finetuning. https://doi.org/10.48550/arXiv.2310.01208 arXiv:2310.01208 [cs].

[26] Philipp Lorenz-Spreen, Lisa Oswald, Stephan Lewandowsky, and Ralph Hertwig. 2023. A systematic review of worldwide causal and correlational evidence on digital media and democracy. *Nature Human Behaviour* 7, 1 (Jan. 2023), 74–101. https://doi.org/10.1038/s41562-022-01460-1

[27] Aamer Madhani and Jill Colvin. 2021. A farewell to @realDonaldTrump, gone after 57,000 tweets. https://apnews.com/article/twitter-donald-trump-ban-cea450b1f12f4ceb8984972a120018d5

[28] Bron Maher. 2024. Twice as many Brits got 2024 election news from TV as from social media, Ofcom finds. https://pressgazette.co.uk/news/ofcom-2024-election-news-sources-television-social-media-working-class/

[29] Aurelia Mohrhoff. 1987. Dalla lingua del Parlamento alla lingua del parlamentare. In *Aurelia Mohrhoff, Dalla lingua del Parlamento alla lingua del parlamentare*. Roma: Camera dei Deputati, 145–156. https://bpr.camera.it/bpr/allegati/show/3815_828_t

[30] North. [n. d.]. French President vs Prime Minister: Who Has More Power? (2026). https://politicahub.com/compare/french-president-vs-french-prime-minister

[31] Benjamin I. Page and Robert Y. Shapiro. 1984. Presidents as Opinion Leaders: Some New Evidence. *Policy Studies Journal* 12, 4 (June 1984), 649–662. https://doi.org/10.1111/j.1541-0072.1984.tb00480.x

[32] Thomas E. Payne. 1997. Describing Morphosyntax: A Guide for Field Linguists. https://doi.org/10.1017/CBO9780511805066 ISBN: 9780511805066.

[33] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. Robust Speech Recognition via Large-Scale Weak Supervision. https://doi.org/10.48550/arXiv.2212.04356 arXiv:2212.04356 [eess].

[34] Robert Riemann. 2026. Synthetic Data | European Data Protection Supervisor. https://www.edps.europa.eu/press-publications/publications/techsonar/synthetic-data

[35] Paulo Rita, Nuno António, and Ana Patrícia Afonso. 2023. Social media discourse and voting decisions influence: sentiment analysis in tweets during an electoral period. *Social Network Analysis and Mining* 13, 1 (March 2023), 46. https://doi.org/10.1007/s13278-023-01048-1

[36] Jan Sawicki, Maria Ganzha, and Marcin Paprzycki. 2023. The State of the Art of Natural Language Processing—A Systematic Automated Review of NLP Literature Using NLP Techniques. *Data Intelligence* 5, 3 (Aug. 2023), 707–749. https://doi.org/10.1162/dint_a_00213

[37] Bruno Castanho Silva and Sven-Oliver Proksch. 2022. Politicians unleashed? Political communication on Twitter and in parliament in Western Europe. *Political Science Research and Methods* 10, 4 (Oct. 2022), 776–792. https://doi.org/10.1017/psrm.2021.36

[38] Alessandro Vinci. 2025. Meloni leader anche sui social, 3,3 milioni di nuovi follower nel 2025: staccato Salvini. Schlein prima per engagement. https://www.corriere.it/tecnologia/25_dicembre_17/meloni-leader-anche-sui-social-3-3-milioni-di-nuovi-follower-nel-2025-staccato-salvini-schlein-prima-per-engagement-431d8cda-a6e3-4d04-880b-922639075xlk.shtml Section: Meloni leader anche sui social, 3,3 milioni di nuovi follower nel 2025: staccato Salvini. Schlein prima per engagement.

Received 01/05/2026